

- There are various factors that affect the distribution and density of population such as physical factors (land forms, vegetation, soils and water supply), climatic factors (temperature, rainfall etc), availability of natural resources, means of transport and communication etc.
- Population growth refers to the change in population. It can be measure in terms of absolute numbers and in percentage. Basic components of population growth are fertility, mortality and migration.
- Migration is the permanent, seasonal or temporary shift of residence for substantial duration.
- Population composition refers to the characteristics of population. These characteristics are measurable and helpful in differentiating one group of people from the other. Age—sex composition, literacy, rural-urban composition, occupation etc are such characteristics.

Demographic Characteristics

- India is one of the most populous country of the world. Ethnic diversity, rural character and uneven distribution etc are some aspects of population affecting the process and pace of socio-economic development of India.
- India's population is unevenly distributed. Plains have more population than the mountains, deserts and forested lands.
- According to 2011 census, India is home to 121.01 crore population. Among states Uttar Pradesh is most populous state in India with population of 19.95 crore followed by Maharashtra, Bihar, West Bengal and Andhra Pradesh. On the other hand, Sikkim shares least proportion of population.
- India's average population density is 382 persons per sq km. Arunachal Pradesh (17) has lowest population density whereas Bihar (1102) has highest density of population. Among Union Territories, Delhi (11297) has highest population density and Andaman and Nicobar (46) has lowest population density.

Growth of Population

- *There are four phases identified for the growth of population in demographic history of India as follow*
 - i. Period of stagnant growth rate (before 1921)
 - ii. Period of steady growth (1921-1951)
 - iii. Period of rapid growth (1951-1981)
 - iv. Period of declining growth rate (after 1981)
- The declining growth rate of population during 2001-2011 was 17.64%. Kerala registered the lowest growth rate of 4.86% whereas Daman and Diu registered the highest growth rate of 53.54%.

Migration

A migrant is one who is enumerated in census at a place other than the place of his birth. In India, heavy pressure of population, poverty, high incidence of unemployment, etc are important factors responsible for migration.

Rural-Urban Composition

India is primarily a country of villages. According to 2011 census, 68.84% of total population lives in rural areas and only 31.16% lives in urban areas, Goa is the most urbanised state where 62.17% of population lives in urban areas. Tamil Nadu (48.45%), Kerala (47.72%), Maharashtra (45.23%) are other states where urbanisation is high. Himachal Pradesh has mostly rural population.

Sex Ratio

Sex ratio refers to the number of females per thousands males. According to 2011 census, India has recorded the sex ratio of 943.

Literacy

The literacy rates among both males and females have shown improvement in census 2011 compared to the last census. The literacy rates in the country as a whole is 74.04%. In the rural and urban areas, the literacy rate are 68.9% and 84.9% respectively. The female literacy rate in the rural and urban areas shows wide variaton. In the urban areas of the country, the female literacy rate is 79.92%, in the rural areas it is only 58.75%.

Tribes of India

- India is the home to large number of indigenous people, who are still untouched by the lifestyle of the modern world. These tribal people also known as adivasis are the poorest in the country, who are still dependent on hunting, agriculture and fishing.
- Some of the major tribal groups in India include Gonds, Santhals, Khasis, Angamis, Bhils, Bhutias and Great Andamanese. All these tribal people have their own culture, tradition, language and lifestyle. This enables the tourist to get an insight into many different cultures at the same time on the tribal tour to India.

Important Tribes of India

- Abors : Arunachal Pradesh
- Aptanis : Arunachal Pradesh
- Badagas : Nilgiri (TN)
- Bhils : Madhya Pradesh and Rajasthan, some in Gujarat and Maharashtra
- Bhot : Himachal Pradesh
- Bhotias : Garhwal and Kumaon regions of Uttar Pradesh
- Chakma : Tripura
- Chenchus : Andhra Pradesh, Orissa
- Gaddis : Himachal Pradesh
- Garos : Meghalaya
- Gonds : Madhya Pradesh. Also in Bihar, Orissa and Andhra Pradesh

- Gujjars : Himachal Pradesh
- Jarawas : Little Andamans
- Khasis : Assam, Meghalaya
- Kol : Madhya Pradesh
- Kotas : Nilgiri (Tamil Nadu)
- Kuki : Manipur
- Lepchas : Sikkim
- Lushais : Mizoram
- Murias : Bastar (Madhya Pradesh)
- Mikirs : Assam
- Mundas : Bihar, Orissa, West Bengal
- Nagas (Angami, Sema, Ao, Tangkul, Lahora): Nagaland, some in Assam and NEFA region.
- Oarons (also called Kurukh): Bihar, Orissa and West Bengal.
- Onges : Andaman and Nicobar islands
- Santals : Birbhum region in Bengal, Hazaribagh, Purnea in Bihar, Orissa
- Sentinelese : Sentinel Island, Andaman and Nicobar
- Shompens : Andaman and Nicobar
- Todas : Nilgiri (Tamil Nadu)
- Uralis : Kerala
- Warlis : Maharashtra

NATURAL HAZARDS AND DISASTER MANAGEMENT

- A disaster is a result of natural or man-made causes that leads to sudden disruption of normal life, causing severe damage to life and property to an extent those available social and economic protection mechanisms are inadequate to cope.
- Disaster management is the managerial function which deals with creating the framework within which communities to reduce vulnerability to hazards and cope with disasters.
- As per origin, disasters can be classified into natural and man-made disasters. Based on generic disasters can be classified into five categories; water and climate, geological, biological, accidental and chemical, industrial and nuclear.

Natural Disaster

It is a major adverse event resulting from natural processes of the Earth; examples include floods, droughts, cyclones, landslides, volcanic eruptions, earthquakes, tsunamis and other geologic processes.

Earthquake

- An earthquake is a vibration or oscillation of the surface of the Earth caused by the elasticity or the isostatic adjustment of the rocks, beneath the surface of the Earth. Major earthquakes are usually caused by sudden movements along faults.
- Earthquakes are by far the most unpredictable and highly destructive of all the natural disasters. There is a geographic-pattern of the earthquake around the world at the tectonic plate margins. Earthquakes that are of tectonic origin have proved to be the most devastating and their area of influence is also quite large.
- **Earthquake Zones of India** On the basis of the intensities or the destructiveness of the earthquakes. India is divided into four earthquake zones.
- Zone IV and Zone V had experienced some of the most devastating earthquakes in India. Areas vulnerable to these earthquakes are the North-East states, areas to the North of Darbhanga and Araria along the Indo-Nepal border in Bihar, Uttarakhand, Western Himachal Pradesh (around Dharamshala) and Kashmir valley in the Himalayan region and the Kutch (Gujarat).
- Zone III covers Southern and South-Eastern parts of Rajasthan, larger parts of Madhya Pradesh, Maharashtra, Karnataka, Jharkhand and Northern and North-Western part of Odisha.

Floods

- These are sudden and temporary inundation of a large area as an overflowing of rivers or reservoirs.
- The common causes which can be responsible for over flowing of river are heavy rainfall, sediment deposition, deforestation, interference in drainage system, change in the course of river etc. Tsunami and cyclone are responsible for flood in coastal areas.

Flood Prone Zones in India

- In India, around 40 million hectare area is flood-prone, which is one eighth of the total area. The most flood prone areas are the Brahmaputra, Ganga and Indus basins.
- As far as states are concerned, Uttar Pradesh, Bihar, West Bengal and Odisha are the most flood affected states followed by Haryana, Punjab and Andhra Pradesh.

Droughts

It is either absence or deficiency of rainfall from its normal pattern in a region for an extended period of time leading to general suffering in the society.

On the basis of severity of droughts, India can be divided into the three regions, which are as follows :

1. **Extreme Drought Affected Areas** Most parts of Rajasthan, Marusthali and Kutch regions of Gujarat fall in this category.
2. **Severe Drought Prone Area** Parts of Eastern Rajasthan, most parts of Madhya Pradesh, Eastern parts of Maharashtra, interior parts of Andhra Pradesh and Karnataka plateau, Northern parts of interior Tamil Nadu and Southern parts of Jharkhand and interior Odisha.

- 3. Moderate Drought Affected Area** Northern parts of Rajasthan, Haryana, Southern districts of Uttar Pradesh, the remaining parts of Gujarat, Maharashtra except Konkan, Jharkhand and Coimbatore plateau of Tamil Nadu and interior Karnataka.

Tsunami

- It is a Japanese word which means Harbor waves. Earthquakes and volcanic eruptions that cause the sea-floor, to move abruptly resulting in sudden displacement of ocean water in the form of high vertical waves are called tsunamis or seismic sea waves.
- Earthquakes, volcanic eruptions and other underwater explosions (including detonations of underwater nuclear devices), landslides, glacier calvings, meteorite impacts and other disturbances above or below water all have the potential to generate a tsunami.
- From the tsunami point of view, Pacific ocean is in the most dangerous position. The tsunami that occurred on the 26th of December, 2004 in the Sumatra island of Indonesia in the Indian ocean was the result of subduction of Indian plate below the Burmese plate.

Landslides

The term landslide includes all varieties of mass movements of hill slopes and can be defined as the downward and outward movement of slope forming materials composed of rocks, soils, artificial fills or combination of all these materials along surface of separation by falling, sliding and flowing, either slowly or quickly from one place to another.

Landslides in India

- Landslide and avalanches are among the major hydro-geological hazards that affect large parts of India. Besides the Himalayas, the North-Eastern hill ranges the Western Ghats, the Nilgiris, the Eastern Ghats and the Vindhyas in that order, covering about 15% of the landmass.
- The Himalayas alone count for landslides of every fame, name and description big and small, quick and creeping, ancient and new. The North-Eastern region is badly affected by landslide problems of a bewildering variety.
- A different variety of landslides, characterised by a lateritic cap, pose constant threat to the Western Ghats in the South, along the steep slopes overlooking the Konkan coast besides Nilgiris, which is highly landslide prone.

Tropical Cyclones

- Tropical cyclones are characterised by destructive winds, torrential rainfall and storm surges disrupt normal life with the accompanying phenomena of floods due to the exceptional level of rainfall and storm surge inundation into inland areas.
- Cyclonic storms are more common in India's Eastern coast and Bangladesh, they routinely hit between April and November, causing deaths and widespread damage to property.

Man-Made Disaster

- Man-made disasters refer to non-natural disastrous occurrences that can be sudden or longer term.
- Sudden man-made disasters include structural collapses, such as building and mine collapse, when this occurs independently without any outside force.

Nuclear Disaster

- With increased emphasis on power generation through nuclear technology, the threat of nuclear hazards has also increased.
- The nuclear facilities in India have adopted internationally accepted guidelines for ensuring safety to the public and environment. A crisis management system is also in place to take care of any nuclear hazard.

Bhopal Gas Tragedy

- It happened in 23rd December, 1984 and considered as world's worst industrial disaster.
- A gas (methyl isocyanate) was leaked from Union Carbide Pesticide plant. This incident cause approximately 2000 deaths.

Disaster Management Act, 2005

- The Disaster Management Act, 2005 has been enacted as the Central Act to deal with the management of disasters. This act envisaged a three tier Disaster Management structure in India at National, States and District levels.
- Under the act, the NDMA, SDMA, NEC, NDRF, NIDM and disaster related funds were established. The Disaster Management Act mandates the Central Government to establish National Disaster Management Authority as nodal authority with Prime Minister as its ex-officio chairperson.
- The DM Act puts on Central Government the obligation to take all measures necessary and expedient for the purpose of disaster management including coordination between ministries and department, state governments, various domestic and international agencies etc. It is also obliged to make proper allocation of funds.

> PRACTICE EXERCISE

1. Which of the following pairs are correctly matched?

- | | |
|-------------------|---|
| 1. Winter rain in | : Western disturbance North-West Indian |
| 2. Summer rain in | : Retreating monsoon Malabar coast |
| 3. Summer rain in | : North-Westerly Bengal basin |
| 4. Winter rain in | : North-East Monsoon Tamil Nadu coast |

Codes

- | | |
|-------------|------------------|
| (a) 1 and 2 | (b) 1, 3 and 4 |
| (c) 3 and 4 | (d) All of these |

2. Amritsar and Shimla are almost on the same latitude, but their climate difference is due to

- (a) the difference in their altitudes
(b) their distance from sea
(c) snowfall in Shimla
(d) pollution in Amritsar

3. Which one of the following is not a causative factor with respect to poor coverage of forest area in Jammu and Kashmir?

- (a) Low amount of rainfall
(b) Large area under cultivation
(c) Steep barren slopes
(d) Snow covered peaks

4. The irregularity in the amount of rainfall in different parts of the North Indian plains is mainly due to

- (a) irregularity intensity of low pressure in the North-Western parts of India
(b) variation in the location of the axis of low pressure trough
(c) difference in frequency of cyclones
(d) variation in the amount of moisture

5. Which of the following regions has the potential for harnessing of tidal energy in India?

- (a) Gulf of Cambay
(b) Gulf of Mannar
(c) Backwaters of Kerala
(d) Chilka Lake

6. Consider the following rivers

- | | |
|----------------|-------------|
| 1. Kishanganga | 2. Ganga |
| 3. Wainganga | 4. Penganga |

The correct sequence of these rivers when arranged in the North-South direction is

- | | |
|----------------|----------------|
| (a) 1, 2, 3, 4 | (b) 2, 1, 3, 4 |
| (c) 2, 1, 4, 3 | (d) 1, 2, 4, 3 |

7. Which one of the following pairs is incorrectly matched?

- | | |
|------------------------|--------------------------------|
| (a) Himalayas tertiary | : Fold mountain |
| (b) Deccan trap | : Volcanic fissure eruption |
| (c) Western Ghat | : Palaeozoic fold, Mountains |
| (d) Aravalli | : Pre-Cambrian relict mountain |

8. The Palk Bay lies between

- (a) Gulf of Kutch and Gulf of Khambhat
(b) Gulf of Mannar and Bay of Bengal
(c) Lakshadweep and Maldiv Islands
(d) Andaman and Nicobar Islands

9. At which one of the following places two important rivers of India originate, while one of them flows towards North and merges with another important river flowing towards Bay of Bengal, the one flows towards Arabian sea?

- (a) Amarkantak (b) Badrinath
(c) Mahabaleshwar (d) Nasik

10. Which of the following pairs is incorrectly matched?

- | | |
|------------------|--------------|
| (a) Imphal basin | : Lacustrine |
| (b) Ladakh plain | : Glacial |
| (c) Konkan coast | : Alluvial |
| (d) Ganga plain | : Alluvial |

11. Identify the incorrect statement about the Karnataka Plateau.

- (a) It has an average elevation of 600-900 m
(b) It is composed of volcanic lava flow of Deccan trap in its Northern part
(c) It has two distinct physiographic features Malnad and Maidan
(d) The highest peak is Kalsubai

12. Which of the following is not correct with respect to Chilka lake?

- (a) It is the largest brackish water lagoon of Asia
(b) It experiences seasonal fluctuations of water level
(c) It is situated South of the Mahanadi Delta
(d) It is situated North of the Mahanadi Delta

13. Which one of the following lakes was formed when the marine transgression had taken place forming a lagoon, but at present is almost a fresh water lake?

- (a) Pulicate (b) Sambhar
(c) Kolleru (d) Vembanad

14. Consider the following statements

1. Rihand dam is on a tributary of the Son river.
2. Hirakud dam is on the Mahanadi river.
3. Tungbhadra project is a joint venture of the Andhra Pradesh and Karnataka states.
4. Kosi is known as the 'Sorrow of Bihar'.

Which of the statements given above are correct?

- (a) 1, 2 and 3 (b) 1, 2 and 4
(c) 1, 3 and 4 (d) All of these

15. Consider the following statement(s)

1. Western Himalayas are very dry whereas Eastern Himalayas are wetter.
2. Western Himalayas rise gradually in a series of ranges whereas Eastern Himalayas rises abruptly.

Which of the statement(s) given above is/are correct?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

16. Which among the following rock system in India is also known as storehouse of minerals?

- (a) Archean rock system
(b) Dharwar system
(c) The Cudappah system
(d) The Vindhyan system

17. With reference to the river Luni, which one of the following statements is correct?

- (a) It flows into Gulf of Khambhat
(b) It flows into Gulf of Kutch
(c) It flows into Pakistan and merges with a tributary of Indus
(d) It is lost in the marshy land of the Rann of Kutch

18. Which of the following is the main stream of the Avadh Plains and its course has been shifting considerably suggesting that it is an aggrading river?

- (a) Kosi (b) Ghaghara
(c) Gandak (d) Chambal

19. Which prominent geomorphic feature separates Shivaliks from the Middle Himalayan Range?

- (a) Fault basins
- (b) Lacustrine basins
- (c) Glacial valleys
- (d) Terai and Bhabar regions

20. Which of the following pairs is incorrectly matched?

Wetland/Lake	State
(a) Pichola	: Rajasthan
(b) Ashtamudi	: Kerala
(c) Uani	: Maharashtra
(d) Kabar	: Uttar Pradesh

21. Consider the following statement(s)

1. Generally, Western ghats are broader than Eastern ghats.
2. Eastern ghats are more continuous than the Western ghats.
3. Eastern ghats act as a water divide.

Which of the following is/are the difference between Eastern and Western Ghats?

- (a) Only 1
- (b) 1 and 2
- (c) 2 and 3
- (d) All of these

22. Which of the following pairs Passes and Locations is/are incorrectly matched?

Passes	Locations
1. Zojila and Burzila	: Jammu and Kashmir
2. Bara Lapcha La	: Uttar Pradesh and Shipki La
3. Thaga La, Niti Pass and Lipu Lekh	: Himachal Pradesh
4. Nathu La and Jelep La	: Sikkim

Codes

- (a) 1 and 2
- (b) 2 and 3
- (c) 3 and 4
- (d) 1, 3 and 4

23. The Amarkantak hill is the source of two rivers flowing in two different directions (West and East). They are

- (a) Narmada and Tapi
- (b) Narmada and Son
- (c) Tapi and Betwa
- (d) Tapi and Son

24. Which of the following is the highest peak of the Zaskar range?

- (a) Kamet
- (b) Dhaulagiri
- (c) Kanchenjunga
- (d) K2

25. The agglomeration of saline depressions on the Western side of the Araval is are generally known as

- (a) Bagar
- (b) Dhrians
- (c) Kallar
- (d) Rann

26. The Amindivi and Cannanore islands are separated from Minicoy island by

- (a) 10° channel
- (b) 9° channel
- (c) 8° channel
- (d) Duncan passage

27. The sand dunes have formed a large number of shallow lagoons along the Malabar coast, these lagoons are generally known as

- (a) Nads
- (b) Kari
- (c) Theris
- (d) Kayals

28. Which one of the following is not true about Rajasthan Bagar?

- (a) It drained by the river Luni
- (b) It has thin blanket of sand
- (c) It has salt lakes
- (d) It has longitudinal and crescent shaped sand dunes

29. What is the most important geographic use of the Himalayas to India?

- (a) Prevention of invasions
- (b) Valuable source of timber
- (c) They prevent the monsoons from crossing to the North and also serve as a reservoir of water from which our perennial rivers flow
- (d) They protect India from the icy cold North winds

30. Which one of the following phenomenon happens when the Sun shines vertically over the Tropic of Capricorn in the Southern hemisphere?

- (a) High pressure develops over North-Western India due to low temperatures
- (b) Low pressure develops over North-Western India due to high temperatures
- (c) No changes in temperature and pressure occur in North-Western India
- (d) 'Loo' blows in the North-Western India

31. Consider the following statements

1. The Garo-Rajmahal gap separates the Meghalaya Plateau from the main block of the Peninsular Plateau.
2. The Karnataka Plateau's hilly region is known as Maland.
3. The backwaters in Kerala are locally known as Kayals.
4. Kolleru lake was once far inland, but now it is a lagoon.

Which of the statements given above are correct?

- (a) 1, 3 and 4
- (b) 1, 2 and 3
- (c) 1, 2 and 4
- (d) All of these

32. Consider the following statements

1. The Western part of Marusthali is covered by shifting sand dunes called Dhrian.
2. The Bhangar lands in Punjab are called as Wetlands.
3. Amarkantak is the highest peak of Satpura range.
4. Vindhya range acts as a watershed between the Ganga system and the river systems of South India.

Which of the statements given above are correct?

- (a) 1, 2 and 3
- (b) 1 and 4
- (c) 2, 3 and 4
- (d) 3 and 4

33. Which one of the following physiographic units has been created by both exogenic and endogenic forces?

- (a) The Peninsular plateau
- (b) The Thar desert
- (c) The Indo-Gangetic plain
- (d) The Himalayas

34. Which of the following statements is correct in respect of India, geographical position?

- (a) Lies between the latitudes of 8° and 37°N
- (b) Lies between the longitudes of 68° and 92°E
- (c) Lies in the North of Equator
- (d) All of the above

35. Which one of the following statements is incorrect?

- (a) Ghaggar's water is utilised in the Indira Gandhi canal
- (b) Narmada rises from Amarkantak region
- (c) Nizam Sagar is situated on the Manjra river
- (d) Penganga is a tributary of the Godavari

36. Consider the following statement(s)

1. The average rainfall in India is about 125 cm.
2. The proportion of drought prone areas to the total area in India is 30%.

Which of the statement(s) given above is/are correct?

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2